18 Open methods, governed coordinates: a FLOSS workflow for responsible LiDAR sharing in the Lower Amazon

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This paper discusses the ethical risks of facilitating looting through open-access data in the Brazilian Amazon. In this region, pre-colonial earthworks and the extent of Amazonian Dark Earth (terra preta) can be identified from satellite imagery, and used to locate sites inside Indigenous, quilombola and other traditional territories. Site exposure invites looting and land speculation. Hes et al. (2026, Nature Ecology & Evolution) argue for systematic ethical consideration of whether and how to reveal what such data contain and conclude that not mapping at all is a scientific option that should always be considered. I take that seriously and pose a FLOSS question: what must exist, in tooling and metadata standards, for mapping to remain defensible more often?

Drawing on a pilot in the Santarém–Belterra region of Pará, I set out a tiered publication model built entirely on free and open-source tools: PDAL and CloudCompare for point clouds, QGIS with the Relief Visualisation Toolbox and Open LiDAR Toolbox for derivation and visualisation, R for analysis, version-controlled and containerised. The workflow separates four layers: (1) code and parameters, fully open; (2) derived terrain and analytical surfaces at generalised resolution, open; (3) site-level geometry, open under a community-agreed licence; (4) precise coordinates of sensitive locations, withheld and released through a named governance process.

I report what each tier costs in reproducibility, how much of the study an independent reader can re-run without layer 4, and use ground-truthing at Sítio Cedro to test whether generalised surfaces stay analytically useful. Current FLOSS geospatial stacks and repository metadata standards have no vocabulary for "open but governed"; supplying one is work for this community. Openness in LiDAR archaeology is not a switch but a gradient, and FLOSS is what makes that gradient implementable.